

Session 2

Machine learning in production

What you build today: Project 1, the ticket triage scoring service.

The job

- Every ticket gets a priority the moment it arrives
- P0 is an outage or a security incident; four percent of tickets
- A missed P0 is an SLA breach. A false P0 costs an engineer's hour
- It ships as a service: version, health, metrics, container

Accuracy is a trap

Model	Accuracy	P0 recall
Always predict P2	0.41	0.00
Naive classifier	high	low
What Northwind needs	lower	high

The metric has to survive contact with the business. Pick it before you train.

Metrics that mean something here

- **Macro-F1**: each class counts equally, so the rare one matters
- **P0 recall**: the commercial number
- **P0 precision**: the price you pay for recall
- **Brier and ECE**: do the probabilities mean what they say?

Reproducible features

- The feature pipeline is a fitted object, serialised with the model
- Same ticket, same vector: today, after restart, in the container
- Train and serve skew is the most common silent failure in deployed ML
- One artifact, one checksum, refused if it does not match

Fix one: class weights

- Tell the loss a P0 mistake costs about ten times a P2 mistake
- Cheap, first thing to try
- Moves the boundary, leaves the probabilities biased

Fix two: a threshold, not argmax

- Predict P0 when its probability clears a value chosen on validation
- The target recall and the precision floor are business numbers
- This is where a product owner can argue with you, and should

Fix three: calibration

- Map scores to probabilities on held-out data
- After it, 0.3 means "three in ten of these are real"
- Before it, a threshold is a threshold on noise

Why a linear model today

- Trains in seconds on a laptop
- Coefficients you can read out loud
- On short text, hard to beat by much
- Session 3 brings the transformer and the harness that measures the difference

Liveness is not readiness

Probe	Question	On failure
/healthz	Is the process up?	Restart the container
/readyz	Can it score right now?	Stop routing traffic

A model that fails to load: alive, not ready, one log line that says why.

The artifact

- `model.joblib` plus `metadata.json` : version, git SHA, data hash, threshold, classes, checksum, metrics
- `latest` is a symlink, not a copy
- The service refuses a corrupt or mixed artifact at startup
- Baked into the image at build; loaded once, not per request

Acceptance criteria

1. Same ticket, same features, before and after reload
2. Macro-F1 clearly above the majority baseline
3. P0 recall at or above 0.8 on the test split, by the threshold rule
4. Artifact versioned, checksummed, reloadable; corrupt artifact refused
5. Service: liveness, readiness, metrics, correlation ID, versioned prediction

`make session02` reports every test passed.

Questions worth asking

- What recall target does a 15 minute P0 SLA imply?
- What happens to the threshold when next quarter's tickets drift?
- What changes in this service if the model becomes a transformer? What stays?